

# Half-Integer Spectral Arithmetic for the Fabius–Rvachev System

Weighted Prouhet cancellation, dyadic  $q$ -binomial aliasing,  
cyclotomic traces, and a Lambert– $W$  endpoint bridge

Frontier research report based on the recursive corpus

Analysis/FabiusFunction/docs

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## Abstract

Let

$$\Phi(z) := \widehat{\text{up}}(z) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{z}{2^j}\right), \quad \text{sinc } u := \frac{\sin(\pi u)}{\pi u},$$

be the Fourier transform of Rvachev's compactly supported up-function, and put  $a_k = \Phi(k + \frac{1}{2})$ . The repository corpus already contains the Fourier product, exact dyadic values of the Fabius function, their Gaussian-binomial and Thue–Morse formulae, the Lambert– $W$  endpoint asymptotic, the complete real-axis Fourier-decay theory assembled from several waves, and a recent spectral account of the integer zero divisor. This report does not relabel those results as new. It develops instead the half-integer spectral sequence  $(a_k)$ , identified in the previous spectral audit as a genuine unresolved bridge.

The first main theorem is an infinite, positive-weight Prouhet system of all orders. With  $o_k = 2k + 1$ ,  $\varepsilon_k = (-1)^{s_2(k)}$ , and  $w_k = |a_k|$ , one has

$$\sum_{k \geq 0} a_k = \frac{1}{2}, \quad \sum_{k \geq 0} \varepsilon_k w_k o_k^{2m} = 0 \quad (m \geq 1).$$

Thus the Thue–Morse partition of the odd integers has exactly matching positive weighted even moments of every order. More generally, the same coefficients form an exact signed quadrature rule  $\sum_k \varepsilon_k w_k P(o_k^2) = P(0)/2$  for every polynomial  $P$ .

The second main theorem is a finite dyadic spectral transform. If  $N = 2^n$  and  $r$  is odd with  $1 \leq r < N$ , define

$$A_{n,r} = \frac{1}{N} \left[ 1 - 2 \sum_{j=1}^{N-1} F\left(\frac{j}{N}\right) \cos\left(\frac{\pi r j}{N}\right) \right].$$

Then the exact alias identity

$$A_{n,r} = \sum_{q \in \mathbb{Z}} \Phi\left(qN + \frac{r}{2}\right)$$

holds, the dyadic samples are recovered by an exact inverse discrete cosine transform, and

$$|A_{n,r} - \Phi(r/2)| \leq \frac{2(1 - 2^{-n})\zeta(n)}{\pi^n} 2^{-n(n-1)/2} \quad (n \geq 2).$$

Substitution of the repository's finite  $q = 1/2$  Gaussian-binomial formula for each  $F(j/N)$  therefore gives superexponentially convergent, finite  $q$ -binomial–Thue–Morse–cyclotomic approximants to every fixed half-integer Fourier amplitude.

The folded coefficients form a Galois orbit in the maximal real  $2^{n+1}$ -st cyclotomic field. Their orbit polynomial lies in  $\mathbb{Q}[X]$ ; all power traces are explicit rational convolutions of dyadic Fabius

values with a Ramanujan-sum congruence filter. These rational traces converge to nonlinear spectral moments  $\sum_k a_k^m$ . An additional exact factorization isolates a finite Thue–Morse polynomial value as the sign carrier of every alias class. Finally, the all-order Prouhet identities turn the endpoint function into an exact “beyond all algebraic orders” spectral remainder. Inserting the repo’s corrected Lambert– $W$  asymptotic produces a direct, explicit bridge between the signed half-integer sinc amplitudes and the lower-Lambert saddle. All new claims below are proved for human verification; none is represented as already formalized in Lean. Novelty is asserted only relative to the reviewed repository corpus and targeted source searches, not as a universal priority claim.

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**Status convention.** **IMPORTED BASELINE.** marks a theorem, identity, or notation already present in the repository or in a cited source. **NEW RESULT PROVED HERE.** marks a statement for which a proof is supplied here and which was not found in the audited TeX corpus or in targeted searches of the cited literature. **OPEN OR CONJECTURAL.** marks conjectures or computational observations. The distinction is essential because the report is designed as a frontier supplement, not a rewrite of the existing corpus.

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## Contents

|          |                                                                       |           |
|----------|-----------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Scope, recursive audit, and the non-duplication boundary</b>       | <b>5</b>  |
| 1.1      | What was inspected . . . . .                                          | 5         |
| 1.2      | Results explicitly treated as baseline . . . . .                      | 5         |
| 1.3      | The gap selected for this report . . . . .                            | 6         |
| 1.4      | Contribution map . . . . .                                            | 6         |
| <b>2</b> | <b>Baseline notation and identities</b>                               | <b>7</b>  |
| 2.1      | The three functions . . . . .                                         | 7         |
| 2.2      | The Fourier product . . . . .                                         | 7         |
| 2.3      | The exact dyadic $q$ -binomial formula . . . . .                      | 8         |
| 2.4      | The lower-Lambert endpoint theorem . . . . .                          | 8         |
| <b>3</b> | <b>The half-integer spectral sequence</b>                             | <b>8</b>  |
| 3.1      | Definition and first values . . . . .                                 | 8         |
| 3.2      | Exact two-periodic cosine expansion . . . . .                         | 9         |
| <b>4</b> | <b>An all-order weighted Prouhet system</b>                           | <b>9</b>  |
| 4.1      | Moment cancellation . . . . .                                         | 9         |
| 4.2      | A nontrivial signed measure with every positive moment zero . . . . . | 10        |
| 4.3      | Certified finite truncations . . . . .                                | 11        |
| <b>5</b> | <b>Dyadic spectral moment filtration</b>                              | <b>11</b> |
| 5.1      | Every derivative as a half-integer spectral moment . . . . .          | 11        |
| 5.2      | Exact termination at reduced dyadic points . . . . .                  | 12        |
| <b>6</b> | <b>An exact beyond-all-orders spectral remainder</b>                  | <b>12</b> |
| 6.1      | Cancellation of every cosine Taylor coefficient . . . . .             | 12        |
| 6.2      | The direct Lambert– $W$ bridge . . . . .                              | 13        |
| <b>7</b> | <b>The exact dyadic alias transform</b>                               | <b>13</b> |
| 7.1      | Folded coefficients . . . . .                                         | 13        |
| 7.2      | Exact inverse transform . . . . .                                     | 14        |

|           |                                                                      |           |
|-----------|----------------------------------------------------------------------|-----------|
| <b>8</b>  | <b>Certified superexponential recovery</b>                           | <b>15</b> |
| 8.1       | A uniform product estimate . . . . .                                 | 15        |
| 8.2       | Numerical convergence for the leading amplitude . . . . .            | 16        |
| <b>9</b>  | <b>Finite <math>q</math>-binomial–cyclotomic approximants</b>        | <b>16</b> |
| 9.1       | Explicit substitution . . . . .                                      | 16        |
| 9.2       | A finite algebraic tower . . . . .                                   | 17        |
| <b>10</b> | <b>Cyclotomic orbit polynomials and rational traces</b>              | <b>17</b> |
| 10.1      | The Galois orbit . . . . .                                           | 17        |
| 10.2      | Ramanujan-filtered trace formula . . . . .                           | 18        |
| 10.3      | Orbit polynomials at the first levels . . . . .                      | 18        |
| 10.4      | Rational traces converging to nonlinear sinc moments . . . . .       | 19        |
| 10.5      | Collapse of the normalized Galois orbit . . . . .                    | 20        |
| <b>11</b> | <b>A finite Thue–Morse sign carrier for every alias class</b>        | <b>20</b> |
| 11.1      | Exact factorization of a folded class . . . . .                      | 20        |
| 11.2      | The finite Thue–Morse polynomial . . . . .                           | 21        |
| 11.3      | A positivity conjecture . . . . .                                    | 22        |
| <b>12</b> | <b>Certified arithmetic of integer-zero jets</b>                     | <b>22</b> |
| 12.1      | The imported local formula . . . . .                                 | 22        |
| 12.2      | Finite exact approximants with a rigorous error . . . . .            | 22        |
| <b>13</b> | <b>Computational verification</b>                                    | <b>23</b> |
| 13.1      | Independent checks performed . . . . .                               | 23        |
| 13.2      | Exact arithmetic workflow . . . . .                                  | 24        |
| 13.3      | A compact reproducibility listing . . . . .                          | 24        |
| <b>14</b> | <b>Lean formalization roadmap</b>                                    | <b>25</b> |
| 14.1      | Layer 1: finite Fourier algebra . . . . .                            | 25        |
| 14.2      | Layer 2: Fourier-series extraction . . . . .                         | 25        |
| 14.3      | Layer 3: alias limits and estimates . . . . .                        | 25        |
| 14.4      | Layer 4: cyclotomic structure . . . . .                              | 25        |
| 14.5      | Layer 5: imported endpoint asymptotics . . . . .                     | 26        |
| 14.6      | Indicative theorem interfaces . . . . .                              | 26        |
| <b>15</b> | <b>Further frontier problems</b>                                     | <b>26</b> |
| 15.1      | Exact signs of the folded coefficients . . . . .                     | 26        |
| 15.2      | Minimal polynomials and stabilizers . . . . .                        | 26        |
| 15.3      | Arithmetic nature of the limiting amplitudes . . . . .               | 26        |
| 15.4      | Spectral moments and nonlinear functionals of up . . . . .           | 27        |
| 15.5      | A genuine transform between the two Gaussian-binomial webs . . . . . | 27        |
| 15.6      | Lambert spectral localization . . . . .                              | 27        |
| 15.7      | A two-variable transform . . . . .                                   | 27        |
| <b>16</b> | <b>Conclusion</b>                                                    | <b>28</b> |
| <b>A</b>  | <b>Corpus inventory</b>                                              | <b>28</b> |
| A.1       | Living documentation: 25 TeX files . . . . .                         | 28        |
| A.2       | Frozen archive: 32 TeX files . . . . .                               | 29        |

|                              |           |
|------------------------------|-----------|
| <b>B Selected source map</b> | <b>31</b> |
|------------------------------|-----------|

# 1 Scope, recursive audit, and the non-duplication boundary

## 1.1 What was inspected

The documentation tree is a stratified research record rather than one article. The same-day recursive audit used for this report contains fifty-seven TeX artifacts: twenty-five living or imported sources and thirty-two frozen historical versions. The living stratum includes the main Fabius–Rvachev exposition, a mathematical glossary, the Lean walkthrough, the non-elementarity study, a large consolidated non-formalized frontier volume, a Thue–Morse formula atlas, a sequence of Fourier-decay reports and audits, and seven source-paper transcriptions. The historical stratum preserves earlier expositions, q-calculus studies, asymptotic investigations, repeated-integration reports, gap registers, and superseded formula atlases. A path ledger is supplied in [appendix A](#).

The current canonical descendants were rechecked separately, because the frontier directory is intentionally consolidated over time: processed drafts may be absorbed into the large canonical volume or removed while their frozen archive remains. Accordingly, the appendix records the audited snapshot; it is not a promise that every historical path remains a live branch-head file after future repository maintenance.

Mathematical extraction was concentrated on the current descendants and on the distinctive claims of each archived line of development. Byte-identical and superseded files were treated as provenance, not as independent evidence of multiple results. The audit was then compared with three frontier reports produced from the same corpus on spectral determinants, spline/quadrature structure, and arithmetic Fourier rays. This prevents accidental duplication of very recent deductions that are not yet part of the main exposition.

## 1.2 Results explicitly treated as baseline

The following are *not* claimed as discoveries here.

1. The relations between the bounded Fabius function  $F$ , its signed global extension  $\mathcal{F}$ , and Rvachev’s even compactly supported function  $\mathfrak{u}$ .
2. The refinement differential equation, the entire Fourier product, the cosine-product form, and rapid decay of  $\Phi$ .
3. Exact rationality at dyadic points, the moment recurrences, the highest-bit recursion, and the finite  $q = 1/2$  Gaussian-binomial–Thue–Morse formula for  $\mathcal{F}(m/2^n)$ .
4. The integer zero set of  $\Phi$ , multiplicity  $1 + \nu_2(m)$  at a nonzero integer  $m$ , strict logarithmic concavity between zeros, and the sign law  $\text{sgn } \Phi(x) = (-1)^{s_2(M)}$  on  $M < x < M + 1$ .
5. The pointwise, metric, probabilistic, extremal, annular, and transfer-operator Fourier-decay results developed in the successive decay reports.
6. The corrected endpoint asymptotic in the lower-Lambert coordinate, including the nonconstant Gamma–zeta periodic fluctuation.
7. The spectral determinant model, spectral zeta and heat trace, all-order integer-zero jets, dyadic q-Euler determinants, and symmetric q-deformation developed in the prior spectral frontier report.
8. Exact rational-ray decay laws, cyclotomic orbit norms, resultants, Ramanujan traces, and the Mersenne half-base bridge developed in the prior arithmetic-ray report.

### 1.3 The gap selected for this report

The previous spectral report ends with the focused problem of classifying the half-integer amplitudes

$$\Phi\left(\frac{2k+1}{2}\right),$$

which govern the leading Taylor coefficients at every integer zero. The main repository exposition contains the exact cosine series in which the same amplitudes reconstruct up, while its dyadic arithmetic chapters contain finite  $q$ -binomial formulae for the values being reconstructed. Yet the reviewed corpus does not develop the finite transform between those two arrays, does not state the all-order weighted Prouhet identities below, and does not connect those cancellations to the corrected Lambert endpoint scale.

This is therefore a natural unfilled intersection of five mature themes already present separately in the repository:

$$\begin{aligned} \text{Thue-Morse signs} &\longleftrightarrow \text{half-integer sinc amplitudes} \\ &\longleftrightarrow \text{dyadic } q\text{-binomial values} \longleftrightarrow \text{cyclotomic traces} \longleftrightarrow \text{Lambert-}W. \end{aligned}$$

### 1.4 Contribution map

[table 1](#) summarizes the boundary between imported material, new deductions proved here, and conjectural extensions.

Table 1: Boundary between imported material and the new theorem package.

| Result                              | Content                                                                                                                                 | Status                                                                    |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Half-integer cosine spectrum        | Isolates the sequence $a_k = \Phi(k + 1/2)$ as the nonzero Fourier coefficients of the two-periodic up-function.                        | <a href="#">IMPORTED BASELINE.</a><br>identity, new organization          |
| Weighted Prouhet theorem            | Proves equality of every positive weighted even moment in the Thue-Morse partition of the odd integers.                                 | <a href="#">NEW RESULT</a><br><a href="#">PROVED HERE.</a>                |
| Spectral quadrature                 | Shows $\sum_k \varepsilon_k w_k P((2k+1)^2) = P(0)/2$ for every polynomial $P$ .                                                        | <a href="#">NEW RESULT</a><br><a href="#">PROVED HERE.</a>                |
| Dyadic moment filtration            | Converts every derivative jet at a reduced dyadic point into a half-integer spectral moment and identifies its exact terminating depth. | <a href="#">NEW RESULT</a><br><a href="#">PROVED HERE.</a><br>formulation |
| Beyond-all-orders remainder         | Rewrites $F(x)$ after subtraction of an arbitrary cosine Taylor polynomial, with all algebraic terms cancelling exactly.                | <a href="#">NEW RESULT</a><br><a href="#">PROVED HERE.</a>                |
| Exact dyadic alias transform        | Gives a finite DCT of dyadic Fabius values equal to a complete folded half-integer alias class.                                         | <a href="#">NEW RESULT</a><br><a href="#">PROVED HERE.</a>                |
| Certified convergence               | Gives a uniform $2^{-n^2/2 + \mathcal{O}(n)}$ error bound from the folded coefficient to $\Phi(r/2)$ .                                  | <a href="#">NEW RESULT</a><br><a href="#">PROVED HERE.</a>                |
| $q$ -binomial spectral approximants | Substitutes the finite Gaussian-binomial formula into the DCT, producing exact cyclotomic approximants.                                 | <a href="#">NEW RESULT</a><br><a href="#">PROVED HERE.</a>                |
| Cyclotomic orbit traces             | Constructs rational orbit polynomials and explicit Ramanujan-filtered power traces.                                                     | <a href="#">NEW RESULT</a><br><a href="#">PROVED HERE.</a>                |
| Spectral power-sum limits           | Shows the rational Galois traces converge to $\sum_k a_k^m$ superexponentially up to a rapidly decreasing tail.                         | <a href="#">NEW RESULT</a><br><a href="#">PROVED HERE.</a>                |
| Thue-Morse sign carrier             | Factors each alias class into a finite root-of-unity Thue-Morse product and an absolutely convergent shifted spectral sum.              | <a href="#">NEW RESULT</a><br><a href="#">PROVED HERE.</a>                |

*Continued on the next page*

Contribution map (continued)

| Result                | Content                                                                                                                               | Status                                 |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Lambert bridge        | Inserts the corrected lower-Lambert asymptotic into the exact Prouhet remainder.                                                      | new consequence of imported asymptotic |
| Alias-sign positivity | Conjectures positivity of the residual shifted spectral sum, equivalent to an exact Thue–Morse sign law for every folded coefficient. | OPEN OR CONJECTURAL.                   |

## 2 Baseline notation and identities

### 2.1 The three functions

Let  $F : [0, 1] \rightarrow [0, 1]$  be the bounded Fabius function. Let

$$\mathcal{F}(x) := \sum_{n=0}^{\infty} \varepsilon_n \operatorname{up}(x - 2n - 1), \quad \varepsilon_n := (-1)^{s_2(n)}, \quad (1)$$

be its signed global extension; the sum is locally finite. On  $[-1, 1]$ ,

$$\operatorname{up}(t) = \mathcal{F}(t + 1) = \begin{cases} F(t + 1), & -1 \leq t \leq 0, \\ F(1 - t) = 1 - F(t), & 0 \leq t \leq 1. \end{cases} \quad (2)$$

The function  $\operatorname{up}$  is even, nonnegative,  $C^\infty$ , supported on  $[-1, 1]$ , normalized by  $\int_{\mathbb{R}} \operatorname{up} = 1$ , and satisfies  $\operatorname{up}(0) = 1$ . Every positive derivative of  $\operatorname{up}$  vanishes at 0, because on the right of zero it is  $1 - F(t)$  and  $F$  is flat at the origin.

The global derivative identities are

$$\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x), \quad \operatorname{up}^{(m)}(t) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m(t + 1)). \quad (3)$$

At nonnegative integers,

$$\mathcal{F}(2b) = 0, \quad \mathcal{F}(2b + 1) = \varepsilon_b. \quad (4)$$

These are repository baseline facts, ultimately descending from Arias de Reyna's formulation [6, 7].

### 2.2 The Fourier product

We use the Fourier convention

$$\widehat{f}(z) = \int_{\mathbb{R}} f(t) e^{-2\pi i t z} dt.$$

Put

$$\Phi(z) := \widehat{\operatorname{up}}(z) = \prod_{j=0}^{\infty} \operatorname{sinc}\left(\frac{z}{2^j}\right), \quad \operatorname{sinc} u = \frac{\sin(\pi u)}{\pi u}. \quad (5)$$

The product is entire and even. For every  $J \geq 1$  and real  $x \neq 0$ , selecting the first  $J$  factors gives the elementary rapid-decay estimate

$$|\Phi(x)| \leq \frac{2^{J(J-1)/2}}{\pi^J |x|^J}. \quad (6)$$

The nonzero real zeros are exactly the integers; at  $m \neq 0$  the multiplicity is  $1 + \nu_2(|m|)$ . If  $M < x < M + 1$ , then

$$\operatorname{sgn} \Phi(x) = (-1)^{s_2(M)}. \quad (7)$$

### 2.3 The exact dyadic $q$ -binomial formula

For later substitution, we record the current corpus formula. With  $(q; q)_n = \prod_{j=1}^n (1 - q^j)$  and the Gaussian binomial coefficient  $\begin{bmatrix} n \\ k \end{bmatrix}_q$ , one has for  $m, n \in \mathbb{N}$  and every rational translation  $c$

$$\begin{aligned} \mathcal{F}\left(\frac{m}{2^n}\right) &= \frac{1}{2^{n^2}(1/2; 1/2)_n} \sum_{k=0}^n \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{2^{k(k-1)}(n+k)!} \\ &\quad \times \sum_{s=0}^{m2^k-1} \varepsilon_s \left(s - m2^k + c\right)^{n+k}. \end{aligned} \quad (8)$$

The right side is independent of  $c$ . When  $0 \leq m \leq 2^n$ , it equals the bounded value  $F(m/2^n)$ . This is **IMPORTED BASELINE**. from the main exposition; the new use below is to insert it into a cyclotomic discrete cosine transform.

### 2.4 The lower-Lambert endpoint theorem

Write  $L = \log 2$  and, for sufficiently small  $x > 0$ ,

$$W = W_{-1}(-Lx), \quad \lambda(x) = -\frac{W}{L}, \quad r(x) = 2^{\lambda(x)} = \frac{\lambda(x)}{x}. \quad (9)$$

Let  $\Psi$  be the smooth one-periodic correction whose nonzero Fourier coefficients are

$$\widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L}, \quad \chi_k = \frac{2\pi i k}{L}. \quad (10)$$

Put

$$\mathfrak{g} = \gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{12}. \quad (11)$$

The corrected endpoint theorem in the corpus states

$$F(x) = \frac{e^{\Psi(-W/L)}}{2^{7/12}\sqrt{\pi x}} \exp\left(\frac{\mathfrak{g} - W - \frac{1}{2}W^2}{L}\right) (1 + \mathcal{O}(\lambda(x)^{-1})). \quad (12)$$

The periodic factor is nonconstant and cannot be deleted from a sharp asymptotic. We shall not reprove (12); instead we derive an exact half-integer spectral identity whose left side is  $F(x)$  and then insert this imported theorem.

## 3 The half-integer spectral sequence

### 3.1 Definition and first values

Define

$$a_k := \Phi\left(k + \frac{1}{2}\right), \quad o_k := 2k + 1, \quad w_k := |a_k|. \quad (13)$$

The sign law (7) gives immediately

$$a_k = \varepsilon_k w_k. \quad (14)$$

The first amplitudes are shown in [table 2](#). Their rapid shrinkage is deceptive: the exact identities below require cancellations over all indices, and no finite initial segment has the moment-vanishing property.



| $k$ | $\varepsilon_k$ | $a_k = \Phi(k + 1/2)$             |
|-----|-----------------|-----------------------------------|
| 0   | +1              | 0.553771275888810095344217416345  |
| 1   | -1              | -0.046202299134306712992574862210 |
| 2   | -1              | -0.008656867928610010895881610116 |
| 3   | +1              | 0.001044808241827280615683472766  |
| 4   | -1              | -0.000356152861629792122112229855 |
| 5   | +1              | 0.000298368336838126419196617073  |
| 6   | +1              | 0.000091404980598947955260616141  |
| 7   | -1              | -0.000006555455730992891195331226 |

Table 2: Initial half-integer amplitudes. The signs are the Thue–Morse word.

### 3.2 Exact two-periodic cosine expansion

Let  $P$  denote the two-periodization of  $\text{up}$ ,

$$P(t) := \sum_{\ell \in \mathbb{Z}} \text{up}(t - 2\ell). \quad (15)$$

Because  $\text{up}$  is supported on  $[-1, 1]$ ,  $P(t) = \text{up}(t)$  for  $-1 \leq t \leq 1$ .

**Proposition 3.1** (Half-integer cosine spectrum). *The Fourier series of  $P$  is absolutely differentiable to every order and is*

$$P(t) = \frac{1}{2} + \sum_{k=0}^{\infty} a_k \cos(\pi o_k t). \quad (16)$$

In particular, on  $[-1, 1]$  the series equals  $\text{up}(t)$ .

*Proof.* The  $m$ th complex Fourier coefficient for period two is

$$c_m = \frac{1}{2} \int_{-1}^1 \text{up}(t) e^{-\pi i m t} dt = \frac{1}{2} \Phi(m/2).$$

For nonzero even  $m$ ,  $m/2$  is a nonzero integer and  $c_m = 0$ . Also  $c_0 = 1/2$ . For  $m = \pm(2k + 1)$ , evenness and reality give  $c_m = a_k/2$ . Thus complex Fourier inversion reduces to (16). Estimate (6) with arbitrarily large  $J$  shows that  $a_k$  is a Schwartz sequence, so the series and every differentiated series converge absolutely and uniformly.  $\square$

**Lean status.** The displayed support-interval identity is already formalized as `Fabius.rvachev_cosine_series` in `PoissonSummation.lean`. That theorem supplies the exact odd half-integer cosine series. The all-orders differentiated-series packaging and the weighted Prouhet, alias-transform, cyclotomic-trace, and Lambert consequences developed below remain separate formalization obligations.

*Remark 3.2.* The half-integer sequence is therefore not an auxiliary sampling choice. It is *exactly* the nonconstant Fourier spectrum of the natural two-periodic extension of  $\text{up}$ . Integer Fourier samples disappear because the entire sinc product vanishes there.

## 4 An all-order weighted Prouhet system

### 4.1 Moment cancellation

The classical Prouhet–Thue–Morse construction balances finitely many *unweighted* power sums on a finite dyadic block. Here the half-integer Fourier amplitudes supply positive, rapidly decreasing weights that balance *every* even power on the infinite set of odd integers.

**Theorem 4.1** (Half-integer weighted Prouhet theorem). *The amplitudes (13) satisfy*

$$\sum_{k=0}^{\infty} a_k = \frac{1}{2}, \quad (17)$$

$$\sum_{k=0}^{\infty} a_k o_k^{2m} = 0 \quad (m \geq 1). \quad (18)$$

Equivalently, for every  $m \geq 1$ ,

$$\boxed{\sum_{\substack{k \geq 0 \\ s_2(k) \text{ even}}} w_k (2k+1)^{2m} = \sum_{\substack{k \geq 0 \\ s_2(k) \text{ odd}}} w_k (2k+1)^{2m}.} \quad (19)$$

All displayed series converge absolutely.

*Proof.* Set  $t = 0$  in (16). Since  $P(0) = \text{up}(0) = 1$ ,

$$1 = \frac{1}{2} + \sum_{k \geq 0} a_k,$$

which proves (17). For  $m \geq 1$ , termwise differentiation gives

$$P^{(2m)}(0) = (-1)^m \pi^{2m} \sum_{k \geq 0} a_k o_k^{2m}.$$

Every positive derivative of  $P = \text{up}$  vanishes at zero, hence the left side is zero. This proves (18). Absolute convergence follows by applying (6) with  $J > 2m + 1$ . Finally substitute  $a_k = \varepsilon_k w_k$  and move the negative-sign terms to the opposite side.  $\square$

**Corollary 4.2** (Exact spectral quadrature at the origin). *For every polynomial  $Q \in \mathbb{C}[X]$ ,*

$$\boxed{\sum_{k=0}^{\infty} \varepsilon_k w_k Q((2k+1)^2) = \frac{Q(0)}{2}.} \quad (20)$$

Thus the signed positive weights  $\varepsilon_k w_k$  evaluate every polynomial in the squared odd variable exactly at the off-grid point 0.

*Proof.* Expand  $Q(X) = \sum_{m=0}^d q_m X^m$  and use (17)–(18).  $\square$

**Remark 4.3** (Why this is not the ordinary finite Prouhet theorem). For a block  $0 \leq k < 2^n$ , the unweighted Thue–Morse signs annihilate powers only below degree  $n$ . In (19), the support is infinite, the nodes are the odd integers, the weights are the absolute half-integer sinc amplitudes, and every even degree is annihilated. The recent Thue–Morse transform literature continues to develop finite Prouhet–Tarry–Escott families [11]; it does not supply the spectral weights appearing here.

## 4.2 A nontrivial signed measure with every positive moment zero

**Corollary 4.4** (Spectral moment-null measure). *The finite signed measure*

$$\nu := \frac{1}{2} \sum_{k=0}^{\infty} a_k (\delta_{o_k} + \delta_{-o_k}) \quad (21)$$

has total mass  $1/2$  and satisfies

$$\int_{\mathbb{R}} x^m d\nu(x) = 0 \quad (m \geq 1). \quad (22)$$

Consequently  $\nu - \frac{1}{2}\delta_0$  is a nonzero finite signed measure whose moments of every nonnegative integer order vanish.

*Proof.* The total variation and every polynomial moment are finite because  $(a_k)$  is Schwartz. Odd moments vanish by symmetry; positive even moments vanish by [theorem 4.1](#); and the mass is [\(17\)](#).  $\square$

*Remark 4.5.* There is no contradiction with uniqueness results for positive moment problems: [\(21\)](#) is signed and has unbounded support. It gives a concrete moment-null object built from the exact spectrum of a positive compactly supported density.

### 4.3 Certified finite truncations

The identities are infinite, but the elementary product bound gives explicit certificates for truncation experiments.

**Proposition 4.6** (Prouhet truncation bound). *Let  $m \geq 0$ ,  $K \geq 1$ , and  $J > 2m + 1$  be integers. Put  $p = J - 2m$ . Then*

$$\left| \sum_{k=K}^{\infty} a_k o_k^{2m} \right| \leq \frac{2^{J(J+1)/2}}{\pi^J} \sum_{k=K}^{\infty} (2k+1)^{-p} \quad (23)$$

$$\leq \frac{2^{J(J+1)/2}}{\pi^J} \left( (2K+1)^{-p} + \frac{(2K+1)^{1-p}}{2(p-1)} \right). \quad (24)$$

For  $m \geq 1$ , the same bound is an a posteriori certificate for the residual of the first  $K$  terms, because the complete sum is zero.

*Proof.* Apply [\(6\)](#) with  $x = o_k/2$ . Since

$$|a_k| \leq \frac{2^{J(J-1)/2}}{\pi^J (k+1/2)^J} = \frac{2^{J(J+1)/2}}{\pi^J o_k^J},$$

multiplication by  $o_k^{2m}$  and summation give [\(23\)](#). For the second line use the monotonicity of  $(2x+1)^{-p}$  and the integral estimate

$$\sum_{k=K}^{\infty} (2k+1)^{-p} \leq (2K+1)^{-p} + \int_K^{\infty} (2x+1)^{-p} dx.$$

$\square$

*Remark 4.7.* The coefficient in [\(23\)](#) records every factor of two from the half-integer substitution  $x = o_k/2$ . The sharper alias estimate in [section 8](#) works directly in the variable  $x$  and therefore uses the constant  $C_K$  of [\(41\)](#) without this substitution.

## 5 Dyadic spectral moment filtration

### 5.1 Every derivative as a half-integer spectral moment

For  $m \geq 0$  define

$$S_m(t) := \sum_{k=0}^{\infty} a_k o_k^m \cos\left(\pi o_k t + \frac{m\pi}{2}\right). \quad (25)$$

**Theorem 5.1** (Spectral derivative identity). *For  $-1 \leq t \leq 1$  and every  $m \geq 0$ ,*

$$S_m(t) = \frac{\text{up}^{(m)}(t)}{\pi^m} = \frac{2^{\binom{m+1}{2}}}{\pi^m} \mathcal{F}(2^m(t+1)). \quad (26)$$

*The series is absolutely and uniformly convergent on  $[-1, 1]$ .*

*Proof.* Differentiate [\(16\)](#)  $m$  times. The derivative of  $\cos(\pi o_k t)$  is  $(\pi o_k)^m \cos(\pi o_k t + m\pi/2)$ . The second equality is the imported global derivative identity [\(3\)](#).  $\square$

## 5.2 Exact termination at reduced dyadic points

**Theorem 5.2** (Dyadic spectral filtration). *Let  $n \geq 1$  and let  $a$  be odd with  $0 < a < 2^n$ . Put  $t = a/2^n$ . Then*

$$S_m(t) = 0 \quad (m > n), \quad (27)$$

$$S_n(t) = -\varepsilon_{(a-1)/2} \frac{2^{\binom{n+1}{2}}}{\pi^n}, \quad (28)$$

$$S_m(t) = \frac{2^{\binom{m+1}{2}}}{\pi^m} \mathcal{F}\left(\frac{2^n + a}{2^{n-m}}\right) \quad (0 \leq m < n). \quad (29)$$

Every value in (29) is rational up to the displayed power of  $\pi$ , and is given explicitly by (8). In particular, the reduced dyadic depth  $n$  is exactly the largest positive order for which the spectral moment  $S_m(t)$  can be nonzero.

*Proof.* By theorem 5.1, the relevant argument of  $\mathcal{F}$  is

$$2^m(t+1) = \frac{2^n + a}{2^{n-m}}.$$

If  $m > n$ , this is an even integer, so (4) gives zero. If  $m = n$ , the argument is the odd integer  $2^n + a = 2b + 1$ , with

$$b = 2^{n-1} + \frac{a-1}{2}.$$

The binary supports of the two summands are disjoint; hence  $\varepsilon_b = -\varepsilon_{(a-1)/2}$ , proving (28). The remaining case is simply the unreduced dyadic argument displayed in (29), to which the global  $q$ -binomial formula applies. The critical value is nonzero, while all larger orders vanish.  $\square$

**Corollary 5.3** (Spectral detection of the denominator). *For an interior reduced dyadic point  $t = a/2^n$  with odd  $a$ ,*

$$n = \max\{m \geq 1 : S_m(t) \neq 0\}. \quad (30)$$

*Thus the denominator depth is encoded intrinsically in the half-integer Fourier spectrum, without inspecting the binary expansion of  $t$ .*

*Remark 5.4.* The underlying finite derivative jet at a dyadic point is baseline material in the repository. The new content is its exact re-expression as a hierarchy of weighted half-integer spectral moments and the resulting denominator detector.

## 6 An exact beyond-all-orders spectral remainder

### 6.1 Cancellation of every cosine Taylor coefficient

For  $M \geq 0$  define

$$\rho_M(y) := 1 - \cos y - \sum_{m=1}^M (-1)^{m+1} \frac{y^{2m}}{(2m)!}, \quad (31)$$

with the sum empty when  $M = 0$ .

**Theorem 6.1** (Exact all-order spectral remainder). *For every  $x \in [0, 1]$  and every integer  $M \geq 0$ ,*

$$F(x) = \sum_{k=0}^{\infty} a_k (1 - \cos(\pi o_k x)), \quad (32)$$

and, more strongly,

$$F(x) = \sum_{k=0}^{\infty} \varepsilon_k w_k \rho_M(\pi o_k x). \quad (33)$$

The series in (33) converges absolutely for every fixed  $M$  and uniformly for  $x \in [0, 1]$ .

*Proof.* For  $0 \leq x \leq 1$ , (2) gives  $F(x) = 1 - \text{up}(x)$ . Subtract (16) from 1 and use  $\sum_k a_k = 1/2$  to obtain (32). Subtracting the first  $M$  nonconstant Taylor terms of  $1 - \cos y$  changes the right side by

$$\sum_{m=1}^M \frac{(-1)^{m+1} (\pi x)^{2m}}{(2m)!} \sum_{k=0}^{\infty} a_k o_k^{2m},$$

which is zero by [theorem 4.1](#). Rapid decay of  $a_k$  dominates the polynomial growth of  $\rho_M(\pi o_k x)$  uniformly on  $[0, 1]$ .  $\square$

*Remark 6.2* (A spectral explanation of flatness). For a fixed individual frequency,  $1 - \cos(\pi o_k x)$  has an ordinary even Taylor series at zero. Formula (33) says that after the frequencies are summed with their exact Thue–Morse sinc weights, *every* algebraic Taylor coefficient disappears. The endpoint value survives only as a collective remainder beyond all finite algebraic orders. This is stronger than checking  $F^{(m)}(0) = 0$  one derivative at a time: the same exact identity holds after subtraction at any prescribed order  $M$ .

## 6.2 The direct Lambert– $W$ bridge

**Corollary 6.3** (Thue–Morse–sinc–Lambert bridge). *Let  $M \geq 0$  be fixed,  $L = \log 2$ , and  $W = W_{-1}(-Lx)$ . As  $x \downarrow 0$ ,*

$$\sum_{k=0}^{\infty} \varepsilon_k w_k \rho_M(\pi o_k x) = \frac{e^{\Psi(-W/L)}}{2^{7/12} \sqrt{\pi x}} \exp\left(\frac{\mathfrak{g} - W - \frac{1}{2}W^2}{L}\right) \times (1 + \mathcal{O}(\lambda(x)^{-1})). \quad (34)$$

The left side is independent of  $M$ , although its individual summands are not.

*Proof.* The left side equals  $F(x)$  exactly by [theorem 6.1](#). Apply the imported endpoint theorem (12).  $\square$

*Remark 6.4* (What has and has not been proved). Equation (34) is a rigorous bridge, but it does not yet identify a sharply localized set of indices  $k$  that carries the asymptotic. The natural lower-Lambert scale  $r(x) = \lambda(x)/x$  strongly suggests such a localization problem. Establishing it requires estimates that preserve the massive cancellations in (33); termwise absolute values would destroy the phenomenon. We formulate this as an open problem in [section 15](#).

## 7 The exact dyadic alias transform

### 7.1 Folded coefficients

Fix  $n \geq 1$  and put  $N = 2^n$ . For every integer  $r$ , define the length- $2N$  folded Fourier coefficient

$$A_{n,r} := \frac{1}{N} \sum_{j=0}^{2N-1} P\left(\frac{j}{N}\right) e^{-\pi i r j / N}. \quad (35)$$

It depends only on  $r \bmod 2N$  and is real because  $P$  is real and even. The nontrivial coefficients are represented by odd  $r$  with  $1 \leq r < N$ .

**Theorem 7.1** (Exact half-integer alias identity). *For every admissible  $n, r$ ,*

$$A_{n,r} = \sum_{q \in \mathbb{Z}} \Phi\left(qN + \frac{r}{2}\right). \quad (36)$$

*The series converges absolutely. Moreover,*

$$A_{n,r} = \frac{1}{N} \left[ 1 + 2 \sum_{j=1}^{N-1} \text{up}\left(\frac{j}{N}\right) \cos\left(\frac{\pi r j}{N}\right) \right] \quad (37)$$

$$= \frac{1}{N} \left[ 1 - 2 \sum_{j=1}^{N-1} F\left(\frac{j}{N}\right) \cos\left(\frac{\pi r j}{N}\right) \right]. \quad (38)$$

*Thus a finite discrete cosine transform of exact dyadic Fabius values equals a complete folded alias class of the infinite sinc product.*

*Proof.* Write the complex Fourier expansion of  $P$  as

$$P(t) = \sum_{m \in \mathbb{Z}} c_m e^{\pi i m t}, \quad c_m = \frac{1}{2} \Phi(m/2).$$

Substitute it in (35). Absolute convergence permits interchange, and the finite character sum is  $2N$  when  $m \equiv r \pmod{2N}$  and zero otherwise. Therefore

$$A_{n,r} = 2 \sum_{m \equiv r \pmod{2N}} c_m = \sum_{q \in \mathbb{Z}} \Phi\left(\frac{r + 2Nq}{2}\right),$$

which is (36). Absolute convergence follows from rapid decay.

Pair the terms  $j$  and  $2N - j$  in (35). The endpoint  $j = N$  contributes zero because  $P(1) = \text{up}(1) = 0$ , while  $P(0) = 1$ ; this gives (37). On  $0 \leq j \leq N$ ,  $\text{up}(j/N) = 1 - F(j/N)$ . Since  $r$  is odd,

$$\sum_{j=1}^{N-1} \cos(\pi r j / N) = 0,$$

which yields (38). □

## 7.2 Exact inverse transform

**Theorem 7.2** (Dyadic spectral reconstruction). *For every integer  $0 \leq j \leq 2N - 1$ ,*

$$P\left(\frac{j}{N}\right) = \frac{1}{2} + \sum_{\substack{1 \leq r < N \\ r \text{ odd}}} A_{n,r} \cos\left(\frac{\pi r j}{N}\right). \quad (39)$$

*In particular, all dyadic values of up at denominator  $N$  are reconstructed from the  $N/2$  folded half-integer classes.*

*Proof.* The normalization in (35) is twice the standard length- $2N$  DFT normalization. Its inverse is therefore

$$P(j/N) = \frac{1}{2} \sum_{r=0}^{2N-1} A_{n,r} e^{\pi i r j / N}.$$

For  $r = 0$ , the alias sum contains  $\Phi(0) = 1$  and only zero-valued nonzero integer samples, so  $A_{n,0} = 1$ . For nonzero even  $r$ , every alias argument is a nonzero integer and  $A_{n,r} = 0$ . Finally  $A_{n,2N-r} = A_{n,r}$ . Pairing the odd frequencies gives (39). □

**Corollary 7.3** (Exact discrete energy identity). *One has*

$$\sum_{\substack{1 \leq r < N \\ r \text{ odd}}} A_{n,r}^2 = \frac{1}{N} \left[ 1 + 2 \sum_{j=1}^{N-1} \text{up} \left( \frac{j}{N} \right)^2 \right] - \frac{1}{2}. \quad (40)$$

*The right side is rational.*

*Proof.* Apply finite Parseval to (39). The constant mode contributes  $1/4$ , and each nonconstant cosine has mean square  $1/2$  on the length- $2N$  grid. Use  $P(0) = 1$ ,  $P(1) = 0$ , and symmetry of the remaining samples.  $\square$

*Remark 7.4* (Continuous limit). As  $n \rightarrow \infty$ , the right side of (40) tends

$$2 \int_0^1 \text{up}(t)^2 dt - \frac{1}{2} = \sum_{k=0}^{\infty} a_k^2,$$

which is also ordinary Fourier Parseval for (16). The finite identity is stronger: before taking a limit it expresses the complete Galois-square trace of the folded spectrum using rational dyadic samples.

## 8 Certified superexponential recovery

### 8.1 A uniform product estimate

**Lemma 8.1** (Selected-factor bound). *For every integer  $K \geq 1$  and real  $x \neq 0$ ,*

$$|\Phi(x)| \leq C_K |x|^{-K}, \quad C_K := \frac{2^{K(K-1)/2}}{\pi^K}. \quad (41)$$

*Proof.* For  $0 \leq j < K$ ,

$$|\text{sinc}(x/2^j)| \leq \frac{2^j}{\pi|x|}.$$

Multiply these  $K$  inequalities and bound every remaining sinc factor by one.  $\square$

**Theorem 8.2** (Uniform aliasing error). *Let  $K > 1$  be an integer. For  $N = 2^n$  and odd  $1 \leq r < N$ ,*

$$|A_{n,r} - \Phi(r/2)| \leq 2C_K N^{-K} (2^K - 1) \zeta(K). \quad (42)$$

*Choosing  $K = n \geq 2$  gives the explicit uniform estimate*

$$|A_{n,r} - \Phi(r/2)| \leq E_n := \frac{2(1 - 2^{-n})\zeta(n)}{\pi^n} 2^{-n(n-1)/2}. \quad (43)$$

*In particular  $E_n = 2^{-n^2/2 + \mathcal{O}(n \log n)}$ , uniformly over all odd  $r < N$ .*

*Proof.* Remove the  $q = 0$  term from (36). For  $q \geq 1$  and  $0 < r < N$ ,

$$|qN \pm r/2| \geq (q - 1/2)N.$$

Hence lemma 8.1 gives

$$\begin{aligned} |A_{n,r} - \Phi(r/2)| &\leq 2C_K N^{-K} \sum_{q=1}^{\infty} (q - 1/2)^{-K} \\ &= 2C_K N^{-K} (2^K - 1) \zeta(K), \end{aligned}$$

using  $\sum_{q \geq 1} (q - 1/2)^{-K} = (2^K - 1) \zeta(K)$ . Substitute  $N = 2^n$  and  $K = n$ ; collecting powers of two gives (43).  $\square$

*Remark 8.3* (Why the rate is so fast). At level  $n$ , the unwanted aliases begin a distance comparable with  $2^n$  from the target. The same sinc product permits an  $n$ -factor power bound, so the alias tail behaves like  $(2^n)^{-n}$  multiplied by the factor  $2^{n(n-1)/2}$  accumulated from dyadic scaling. The net exponent is  $-n(n-1)/2$ .

## 8.2 Numerical convergence for the leading amplitude

The distinguished coefficient  $A_{n,1}$  converges to

$$h := \Phi(1/2) = 0.553771275888810095344217416344548881284886642554 \dots \quad (44)$$

The finite values in [table 3](#) were computed from exact rational  $F(j/2^n)$  values, with only the final cyclotomic evaluation performed at high precision.

| $n$ | $A_{n,1}$                   | actual error            | rigorous $E_n$          |
|-----|-----------------------------|-------------------------|-------------------------|
| 2   | 0.5544487530108746285614747 | $6.7748 \cdot 10^{-4}$  | $1.2500 \cdot 10^{-1}$  |
| 3   | 0.5537613501982070710846451 | $9.9257 \cdot 10^{-6}$  | $8.4805 \cdot 10^{-3}$  |
| 4   | 0.5537712795003417162450708 | $3.6115 \cdot 10^{-9}$  | $3.2552 \cdot 10^{-4}$  |
| 5   | 0.5537712758798224271695171 | $8.9877 \cdot 10^{-12}$ | $6.4112 \cdot 10^{-6}$  |
| 6   | 0.5537712758888101658815125 | $7.0537 \cdot 10^{-17}$ | $6.3578 \cdot 10^{-8}$  |
| 7   | 0.5537712758888100953109457 | $3.3272 \cdot 10^{-20}$ | $3.1590 \cdot 10^{-10}$ |
| 8   | 0.5537712758888100953442174 | $5.0949 \cdot 10^{-27}$ | $7.8534 \cdot 10^{-13}$ |
| 9   | 0.5537712758888100953442174 | $4.8386 \cdot 10^{-31}$ | $9.7639 \cdot 10^{-16}$ |
| 10  | 0.5537712758888100953442174 | $1.3837 \cdot 10^{-39}$ | $6.0700 \cdot 10^{-19}$ |

Table 3: Superexponential recovery of the first half-integer amplitude. The bound is deliberately elementary and uniform in  $r$ , so it is not expected to track the much smaller observed error closely.

## 9 Finite $q$ -binomial–cyclotomic approximants

### 9.1 Explicit substitution

The alias transform is finite, while every dyadic value in it has the finite Gaussian-binomial form (8). Their composition is the promised bridge between the two coefficient webs.

**Theorem 9.1** ( $q$ -binomial half-integer approximant). *Let  $N = 2^n$  and let  $r$  be odd with  $1 \leq r < N$ . Then*

$$A_{n,r} = \frac{1}{N} - \frac{2}{N 2^{n^2}(1/2; 1/2)_n} \sum_{j=1}^{N-1} \cos\left(\frac{\pi r j}{N}\right) \times \sum_{k=0}^n \frac{[n]_{1/2}}{2^{k(k-1)}(n+k)!} \sum_{s=0}^{j2^k-1} \varepsilon_s (s - j2^k)^{n+k}. \quad (45)$$

For every fixed odd  $r$ , the right side converges to  $\Phi(r/2)$  as  $n \rightarrow \infty$  through levels with  $r < 2^n$ , and the error is at most  $E_n$ .

*Proof.* Insert (8) with  $m = j$  and  $c = 0$  into (38). The convergence statement is [theorem 8.2](#).  $\square$

*Remark 9.2* (Three structures in one finite expression). Formula (45) simultaneously contains

1. Gaussian binomial coefficients at  $q = 1/2$ ;
2. finite Thue–Morse power sums;



3. real  $2^{n+1}$ -st cyclotomic numbers  $\cos(\pi r j / 2^n)$ .

No infinite product or numerical quadrature is needed to construct the approximant. The infinite sinc product enters only as the certified limit.

## 9.2 A finite algebraic tower

Let

$$\zeta_{2N} = e^{\pi i / N}, \quad K_n := \mathbb{Q}(\zeta_{2N} + \zeta_{2N}^{-1}). \quad (46)$$

Since every  $F(j/N)$  is rational,

$$A_{n,r} \in K_n. \quad (47)$$

The fields form the nested real dyadic cyclotomic tower  $K_1 \subset K_2 \subset \dots$ . The approximants therefore approach the transcendence-unknown numbers  $\Phi(r/2)$  through an explicit tower of exact algebraic numbers with a superexponential analytic certificate.

*Remark 9.3.* The phrase “transcendence-unknown” is intentional. No irrationality or transcendence assertion for  $h = \Phi(1/2)$  is proved here or in the reviewed corpus. The algebraic approximants are useful precisely because they separate exact finite arithmetic from the unresolved nature of the limiting constants.

# 10 Cyclotomic orbit polynomials and rational traces

## 10.1 The Galois orbit

The real cyclotomic Galois group is

$$G_n \simeq (\mathbb{Z}/2N\mathbb{Z})^\times / \{\pm 1\}. \quad (48)$$

Its elements may be represented by odd integers  $u$  modulo  $2N$ , with  $u$  and  $-u$  identified. The representatives  $1 \leq r < N$ ,  $r$  odd, form a complete set.

**Proposition 10.1** (Galois action). *If  $\sigma_u(\zeta_{2N}) = \zeta_{2N}^u$ , then*

$$\sigma_u(A_{n,r}) = A_{n,r'}, \quad (49)$$

where  $r'$  is the unique odd representative in  $[1, N)$  congruent to  $\pm ur \pmod{2N}$ .

*Proof.* Apply  $\sigma_u$  to (38). It fixes the rational Fabius values and sends  $\cos(\pi r j / N)$  to  $\cos(\pi u r j / N)$ . Reduction modulo  $2N$  and evenness of cosine give the stated representative.  $\square$

**Definition 10.2** (Orbit polynomial and power traces). Define

$$\mathcal{P}_n(X) := \prod_{\substack{1 \leq r < N \\ r \text{ odd}}} (X - A_{n,r}) \quad (50)$$

and, for  $m \geq 1$ ,

$$p_m(n) := \sum_{\substack{1 \leq r < N \\ r \text{ odd}}} A_{n,r}^m. \quad (51)$$

The polynomial is called an orbit polynomial rather than a minimal polynomial: if the stabilizer of  $A_{n,1}$  is nontrivial, repeated conjugates may occur.

**Theorem 10.3** (Rational orbit polynomial). *For every  $n \geq 1$ ,*

$$\mathcal{P}_n(X) \in \mathbb{Q}[X], \quad p_m(n) \in \mathbb{Q} \quad (m \geq 1). \quad (52)$$

*The coefficients of  $\mathcal{P}_n$  are recovered from the  $p_m(n)$  by Newton’s identities.*

*Proof.* The multiset of roots is invariant under the full Galois group by [proposition 10.1](#); hence every elementary symmetric function is fixed by  $G_n$  and lies in  $\mathbb{Q}$ . The same argument applies to power sums. Newton's identities are purely rational algebra.  $\square$

## 10.2 Ramanujan-filtered trace formula

Put

$$P_j := P(j/N), \quad 0 \leq j < 2N. \quad (53)$$

Every  $P_j$  is rational, since it is a dyadic value of  $\text{up}$ .

**Theorem 10.4** (Exact rational power traces). *For every  $m \geq 1$ ,*

$$p_m(n) = \frac{1}{2} N^{1-m} \sum_{\substack{0 \leq j_1, \dots, j_m < 2N \\ N \mid j_1 + \dots + j_m}} (-1)^{(j_1 + \dots + j_m)/N} \prod_{\ell=1}^m P_{j_\ell}. \quad (54)$$

*In particular,*

$$p_1(n) = \frac{1}{2} \quad (n \geq 1). \quad (55)$$

For  $m = 2$ , (54) reduces to the discrete energy identity (40).

*Proof.* Expand  $A_{n,r}^m$  from (35) and sum over the  $N/2$  odd representatives. Since  $A_{n,r} = A_{n,2N-r}$ , the representative sum is one half of the sum over all odd residues modulo  $2N$ . Thus

$$p_m(n) = \frac{1}{2N^m} \sum_{0 \leq j_1, \dots, j_m < 2N} \left( \prod_{\ell=1}^m P_{j_\ell} \right) c_{2N}(j_1 + \dots + j_m),$$

where

$$c_{2N}(S) = \sum_{\substack{r \bmod 2N \\ r \text{ odd}}} e^{-\pi i r S / N}$$

is the Ramanujan sum for the power-of-two modulus  $2N$ . Its elementary value is

$$c_{2N}(S) = \begin{cases} N(-1)^{S/N}, & N \mid S, \\ 0, & N \nmid S. \end{cases} \quad (56)$$

Substitution gives (54). For  $m = 1$ , the only indices with  $N \mid j$  are  $j = 0, N$ , so  $p_1(n) = \frac{1}{2}(P_0 - P_N) = 1/2$ .  $\square$

*Remark 10.5* (A trace that does not disperse). The orbit degree is  $N/2 \rightarrow \infty$ , yet the unnormalized first trace remains exactly  $1/2$ . Most conjugates become tiny; a sparse collection of low-frequency conjugates carries the persistent trace. This is quantified below.

## 10.3 Orbit polynomials at the first levels

At  $n = 1$  there is one coefficient,  $A_{1,1} = 1/2$ , so

$$\mathcal{P}_1(X) = X - \frac{1}{2}. \quad (57)$$

At  $n = 2$ ,

$$A_{2,1} = \frac{1}{4} + \frac{31\sqrt{2}}{144}, \quad A_{2,3} = \frac{1}{4} - \frac{31\sqrt{2}}{144}, \quad (58)$$

and

$$\mathcal{P}_2(X) = X^2 - \frac{1}{2}X - \frac{313}{10368}. \quad (59)$$

At  $n = 3$ , direct use of (54) and Newton identities gives

$$\mathcal{P}_3(X) = X^4 - \frac{1}{2}X^3 - \frac{1627}{55296}X^2 - \frac{4603}{23887872}X + \frac{32257}{220150628352}. \quad (60)$$

The corresponding rational power traces are

$$p_1(3) = \frac{1}{2}, \quad p_2(3) = \frac{8539}{27648}, \quad (61)$$

$$p_3(3) = \frac{1351363}{7962624}, \quad p_4(3) = \frac{5175727235}{55037657088}. \quad (62)$$

The four conjugates are approximately

$$\begin{aligned} &0.553761350198207071, & -0.046098883415321902, \\ &-0.008349869595552727, & 0.000687402812667557. \end{aligned} \quad (63)$$

They already resemble the first four half-integer amplitudes, but each includes all aliases in its congruence class.

At  $n = 4$  the eight conjugates are approximately

$$\begin{aligned} &0.553771279500341716, & -0.046202534866953151, \\ &-0.008657957107189789, & 0.001046027211381176, \\ &-0.000358624398713618, & 0.000308087511637062, \\ &0.000103651451631249, & -0.000009929302134645. \end{aligned}$$

Their sum is exactly  $1/2$ .

#### 10.4 Rational traces converging to nonlinear sinc moments

Define the nonlinear half-integer spectral sums

$$\mathfrak{s}_m := \sum_{k=0}^{\infty} a_k^m \quad (m \geq 1). \quad (64)$$

They converge absolutely. The case  $m = 1$  is  $\mathfrak{s}_1 = 1/2$ ; for  $m = 2$ ,

$$\mathfrak{s}_2 = 2 \int_0^1 \text{up}(t)^2 dt - \frac{1}{2}. \quad (65)$$

**Theorem 10.6** (Cyclotomic trace limits). *For every fixed integer  $m \geq 1$ ,*

$$\lim_{n \rightarrow \infty} p_m(n) = \mathfrak{s}_m. \quad (66)$$

*For  $m = 1$  the equality is exact at every level. For fixed  $m \geq 2$ , the error splits into*

$$|p_m(n) - \mathfrak{s}_m| \leq \frac{Nm}{2}(1 + E_n)^{m-1}E_n + \sum_{k \geq N/2} |a_k|^m, \quad (67)$$

*where both terms decay faster than every exponential in  $n$  after applying an  $n$ -factor sinc bound to the tail.*

*Proof.* Index the admissible residues by  $r = 2k + 1$ ,  $0 \leq k < N/2$ . Then [theorem 8.2](#) gives  $|A_{n,2k+1} - a_k| \leq E_n$ . Since  $|a_k| \leq 1$  and  $E_n \leq 1/8$  for  $n \geq 2$ , the mean-value inequality for  $x \mapsto x^m$  gives

$$|A_{n,2k+1}^m - a_k^m| \leq m(1 + E_n)^{m-1} E_n.$$

Summing over  $k < N/2$  and adding the omitted tail proves (67). The first term is superexponential because  $NE_n = 2^{-n^2/2 + \mathcal{O}(n)}$ . For the tail, use [lemma 8.1](#) with  $K = n$ :

$$|a_k| \leq C_n(k + 1/2)^{-n}.$$

The resulting integral bound is

$$\mathcal{O}(C_n^m (N/2)^{1-nm}) = 2^{-mn^2/2 + \mathcal{O}(n)},$$

where the implied constant may depend on the fixed exponent  $m$ . □

**Corollary 10.7** (Algebraic approximation to spectral norms). *Every  $\mathfrak{s}_m$  is a limit of explicitly computable rational numbers  $p_m(n)$  whose defining algebraic conjugates are given by the finite  $q$ -binomial formula (45). In particular,  $2 \int_0^1 \text{up}^2 - 1/2$  admits a superexponentially convergent rational sequence coming from cyclotomic Galois square traces.*

## 10.5 Collapse of the normalized Galois orbit

Let

$$\mu_n := \frac{2}{N} \sum_{\substack{1 \leq r < N \\ r \text{ odd}}} \delta_{A_{n,r}}. \quad (68)$$

This is a probability measure on  $\mathbb{R}$ .

**Theorem 10.8** (Galois-orbit collapse). *The measures  $\mu_n$  converge weakly to the point mass  $\delta_0$ . More quantitatively,*

$$\int x^2 d\mu_n(x) = \frac{2}{N} p_2(n) \leq \frac{3}{N}. \quad (69)$$

Nevertheless the unnormalized trace remains  $p_1(n) = 1/2$ .

*Proof.* From (40) and  $0 \leq \text{up} \leq 1$ ,

$$p_2(n) \leq \frac{1}{N}(1 + 2(N-1)) - \frac{1}{2} < \frac{3}{2}.$$

This proves (69). Chebyshev's inequality shows that the proportion of conjugates outside any fixed neighborhood of zero tends to zero, hence  $\mu_n \Rightarrow \delta_0$ . □

*Remark 10.9.* This is an unusual arithmetic picture: the degree of the orbit grows, the normalized orbit collapses to zero, but its unnormalized first trace remains fixed. The low-index half-integer amplitudes survive as a vanishing proportion of the conjugates and carry the entire trace in the limit.

## 11 A finite Thue–Morse sign carrier for every alias class

### 11.1 Exact factorization of a folded class

Let  $N = 2^n$ , let  $r$  be odd with  $1 \leq r < N$ , and put

$$\alpha := \frac{r}{2N} \in (0, 1/2). \quad (70)$$

Define the finite prefactor

$$\Gamma_{n,r} := \frac{\prod_{h=1}^n \sin(\pi r/2^h)}{\pi^n 2^{n(n+1)/2}}. \quad (71)$$

**Theorem 11.1** (Alias-class factorization). *For every  $q \in \mathbb{Z}$ ,*

$$\Phi\left(qN + \frac{r}{2}\right) = \Gamma_{n,r} \frac{\Phi(q + \alpha)}{(q + \alpha)^n}. \quad (72)$$

Consequently

$$A_{n,r} = \Gamma_{n,r} \sum_{q \in \mathbb{Z}} \frac{\Phi(q + \alpha)}{(q + \alpha)^n}. \quad (73)$$

The shifted sum converges absolutely.

*Proof.* Split the product (5) after the first  $n$  factors. Since  $qN + r/2 = N(q + \alpha)$ , the tail is

$$\prod_{j=n}^{\infty} \operatorname{sinc}\left(\frac{qN + r/2}{2^j}\right) = \Phi(q + \alpha).$$

For  $0 \leq j < n$ ,

$$\sin\left(\pi \frac{qN + r/2}{2^j}\right) = \sin\left(\pi q 2^{n-j} + \frac{\pi r}{2^{j+1}}\right) = \sin\left(\frac{\pi r}{2^{j+1}}\right),$$

because  $q 2^{n-j}$  is even. The denominators of the first  $n$  sinc factors contribute

$$\prod_{j=0}^{n-1} \frac{2^j}{\pi N(q + \alpha)} = \frac{1}{\pi^n 2^{n(n+1)/2} (q + \alpha)^n}.$$

Multiplication proves (72); summing over  $q$  and using (36) gives (73).  $\square$

## 11.2 The finite Thue–Morse polynomial

Let

$$T_n(z) := \prod_{\ell=0}^{n-1} (1 - z^{2^\ell}) = \sum_{m=0}^{2^n-1} \varepsilon_m z^m. \quad (74)$$

**Proposition 11.2** (Root-of-unity magnitude and sign). *For odd  $1 \leq r < N$ ,*

$$\left| T_n\left(e^{2\pi i r/N}\right) \right| = 2^n \left| \prod_{h=1}^n \sin(\pi r/2^h) \right|, \quad (75)$$

$$\operatorname{sgn} \Gamma_{n,r} = (-1)^{\sum_{h=1}^n \lfloor r/2^h \rfloor} = \varepsilon_{(r-1)/2}. \quad (76)$$

Thus the finite prefactor in (73) carries exactly the Thue–Morse sign attached to the target amplitude  $\Phi(r/2)$ .

*Proof.* For the magnitude, use  $|1 - e^{2\pi i x}| = 2|\sin \pi x|$  in each factor of (74) and reverse the order of scales. For the sign,  $\sin(\pi x)$  has sign  $(-1)^{\lfloor x \rfloor}$  away from integers. Since  $r < N$ , all floors beyond  $h = n$  vanish, and

$$\sum_{h=1}^n \left\lfloor \frac{r}{2^h} \right\rfloor = r - s_2(r).$$

Write  $r = 2c + 1$ . Then  $s_2(r) = s_2(c) + 1$ , so  $r - s_2(r) \equiv s_2(c) \pmod{2}$ . Hence the sign is  $\varepsilon_c$  with  $c = (r - 1)/2$ .  $\square$

*Remark 11.3.* The earlier arithmetic-ray report uses root-of-unity values of finite Thue–Morse polynomials to control products along rational doubling orbits. The mechanism here is different: a finite root-of-unity product is extracted from a *Poisson alias class* of the sinc product. The same automatic sign word therefore governs both real-axis intervals and finite dyadic spectral folding.

### 11.3 A positivity conjecture

Computations through  $n = 7$  show

$$\operatorname{sgn} A_{n,r} = \varepsilon_{(r-1)/2} \quad (77)$$

for every admissible  $r$ . By (76), this is equivalent to positivity of the residual shifted sum.

**Conjecture 11.4** (Shifted spectral positivity). *For every  $n \geq 1$  and odd  $1 \leq r < 2^n$ , with  $\alpha = r/2^{n+1}$ ,*

$$\sum_{q \in \mathbb{Z}} \frac{\Phi(q + \alpha)}{(q + \alpha)^n} > 0. \quad (78)$$

*Equivalently, every folded coefficient has the exact Thue–Morse sign (77).*

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*Remark 11.5* (Why the conjecture is nontrivial). The summands in (78) have mixed signs. Neither the positivity of up nor the elementary alias error bound proves the claim, because for residues near  $N$  the target amplitude can be much smaller than the uniform alias bound. A proof may require a quasi-periodic Green-kernel representation or a total-positivity property of the dyadic sample sequence.

## 12 Certified arithmetic of integer-zero jets

### 12.1 The imported local formula

Let  $m \neq 0$  be an integer and write

$$m = 2^v q, \quad q \text{ odd}, \quad d = v + 1. \quad (79)$$

The spectral determinant report proves the exact leading-jet identity

$$\Phi^{(d)}(m) = -\frac{d!}{m^d} \Phi(q/2). \quad (80)$$

It also gives every higher coefficient, but the leading term already shows why the half-integer amplitudes are arithmetically central: every nonzero integer zero reduces to one odd half-integer value.

### 12.2 Finite exact approximants with a rigorous error

**Theorem 12.1** ( $q$ -binomial approximation of every leading zero jet). *Let  $m = 2^v q \neq 0$  with  $q$  odd and  $d = v + 1$ . Choose  $n \geq 2$  with  $2^n > |q|$ , and put  $r = |q|$ . Define*

$$J_{m,n} := -\frac{d!}{m^d} A_{n,r}. \quad (81)$$

*Then  $J_{m,n}$  is an explicitly computable element of the real dyadic cyclotomic field  $K_n$ , given by the finite  $q$ -binomial expression (45), and*

$$|\Phi^{(d)}(m) - J_{m,n}| \leq \frac{d!}{|m|^d} E_n. \quad (82)$$

Thus every leading integer-zero jet has a superexponentially convergent sequence of exact algebraic approximants.

*Proof.* Evenness gives  $\Phi(q/2) = \Phi(r/2)$ . Combine the imported identity (80) with theorem 8.2 and multiply the error by  $d!/|m|^d$ .  $\square$

**Example 12.2** (The zero at  $m = 12$ ). Here  $12 = 2^2 \cdot 3$ , so  $d = 3$  and

$$\Phi^{(3)}(12) = -\frac{6}{12^3} \Phi(3/2) = \frac{-\Phi(3/2)}{288}.$$

Using the level-4 folded coefficient  $A_{4,3} = -0.04620253486695315055 \dots$  gives

$$J_{12,4} = 0.000160425468288031773 \dots,$$

while the exact target is

$$\Phi^{(3)}(12) = 0.000160424649771898309 \dots$$

The rigorous error follows immediately from (82). Increasing  $n$  replaces the displayed algebraic number by a much sharper one without changing the proof architecture.

*Warning 12.3.* The final decimal in the target value in example 12.2 is included only as a numerical orientation. The certified content is the exact formula and the bound (82); the report's reproducibility code computes as many digits as desired.

## 13 Computational verification

### 13.1 Independent checks performed

The numerical work was designed to check identities at several logically independent interfaces rather than merely reproduce one formula in two notations. The following were evaluated using exact rational arithmetic whenever possible and 80–100 decimal digits for the final transcendental operations.

1. The moment recurrence for  $c_k$  and the closed moment–Thue–Morse formula for  $F(a/2^n)$ , including  $F(1/4) = 5/72$ ,  $F(1/8) = 1/288$ ,  $F(3/8) = 73/288$ , and  $F(5/16) = 305857/2073600$ .
2. Direct sinc-product values of  $a_k$  against the cosine reconstruction of up.
3. The weighted moment residuals  $\sum_{k < K} a_k o_k^{2m}$  for increasing  $K$ , together with rapid-decay tail bounds.
4. The exact DCT formula (38) against direct alias summation (36).
5. The convergence table table 3 and the rigorous bound (43).
6. The Galois conjugates at levels  $n = 2, 3, 4$ , their rational power traces, and the orbit polynomials (59)–(60).
7. The single-alias factorization (72) for positive and negative alias indices.
8. The observed sign law (77) for every odd residue at levels  $1 \leq n \leq 7$ .
9. The leading zero-jet approximation (82) for integers of several different dyadic valuations.

### 13.2 Exact arithmetic workflow

The most stable exact path uses the even moments of up. Set  $c_0 = 1$  and compute

$$(2m+1)(2^{2m}-1)c_m = \sum_{j=0}^{m-1} \binom{2m+1}{2j} c_j. \quad (83)$$

Then the repository's closed formula is

$$F\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{a-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} (2a-2h-1)^{n-2k} c_k. \quad (84)$$

Every quantity in (84) is rational. It is then inserted into (38); exact algebraic simplification is needed only for the cosine values.

### 13.3 A compact reproducibility listing

Listing 1: Core exact computation used for the tables.

```
from fractions import Fraction
from math import comb, factorial

def tm_sign(k: int) -> int:
    return -1 if k.bit_count() % 2 else 1

def up_even_moments(max_k: int) -> list[Fraction]:
    c = [Fraction(1)]
    for m in range(1, max_k + 1):
        numerator = sum(
            Fraction(comb(2*m + 1, 2*j)) * c[j]
            for j in range(m)
        )
        denominator = (2*m + 1) * (2**(2*m) - 1)
        c.append(numerator / denominator)
    return c

def fabius_dyadic(a: int, n: int, c: list[Fraction]) -> Fraction:
    total = Fraction(0)
    for h in range(a):
        inner = sum(
            Fraction(comb(n, 2*k) * (2*a - 2*h - 1)**(n - 2*k))
            * c[k]
            for k in range(n//2 + 1)
        )
        total += tm_sign(h) * inner
    return total / Fraction(2**(n*(n + 1)//2) * factorial(n))

# A_{n,r} = 1/N - (2/N) sum_j F(j/N) cos(pi*r*j/N).
# Use exact Fraction values for F and a high-precision cyclotomic backend
# (or symbolic radicals for small n) for the cosine evaluation.
```

*Remark 13.1 (Reproducibility versus proof).* The computations are checks, not substitutes for the proofs. Every central identity in the report is established symbolically before numerical evidence is invoked. The code is included to make constants, orbit polynomials, and error tables independently reproducible.



## 14 Lean formalization roadmap

The new theorem package separates naturally into finite algebra, elementary analysis, and imported high-level asymptotics. A low-risk formalization order is as follows.

### 14.1 Layer 1: finite Fourier algebra

Formalize first the length- $2N$  DFT identities on a finite cyclic group:

1. orthogonality of characters modulo  $2N$ ;
2. vanishing of all nonzero even folded modes;
3. the inverse cosine transform (39);
4. finite Parseval and (40);
5. the power-of-two Ramanujan sum (56);
6. the rational trace formula (54).

This layer uses only finite sums, rational dyadic values already available in the project, roots of unity, and elementary complex algebra.

### 14.2 Layer 2: Fourier-series extraction

Use the established rapid decay of  $\widehat{u_P}$  to formalize:

1. the periodization  $P$  and its complex Fourier coefficients;
2. the reduction to odd modes and the cosine series (16);
3. termwise differentiation to every order;
4. the weighted Prouhet identities and polynomial quadrature;
5. the derivative filtration (27)–(29);
6. the exact spectral remainder (33).

The main analytic input is a reusable theorem that a Schwartz coefficient sequence permits uniform termwise differentiation of a Fourier series.

### 14.3 Layer 3: alias limits and estimates

Formalize the infinite alias identity by applying the finite character filter to an absolutely convergent Fourier series. Then prove the selected-factor bound and sum the shifted zeta tail. The exact target statements are (36), (42), and (43). No contour integration is needed.

### 14.4 Layer 4: cyclotomic structure

Two implementation choices are possible.

1. Work concretely in  $\mathbb{C}$  with a primitive root  $\zeta_{2N}$  and prove invariance of symmetric polynomials under the explicit automorphisms.
2. Work in an abstract cyclotomic extension, use the maximal real subfield only at the interface, and obtain rationality from fixed-field lemmas.

The first route is likely shorter for the DCT formula; the second gives cleaner Galois traces. The orbit polynomial should be introduced without asserting it is minimal.

## 14.5 Layer 5: imported endpoint asymptotics

Once (33) is formalized, the Lambert bridge is a one-line rewrite using the project's existing endpoint theorem. This is much easier than formalizing a new saddle analysis. A later project can address the genuinely new localization problem posed in [conjecture 15.5](#).

## 14.6 Indicative theorem interfaces

Listing 2: Schematic Lean theorem signatures.

```
-- Half-integer weighted Prouhet cancellation.
theorem halfIntegerSpectrum_evenMoment_zero (m : Nat) (hm : 0 < m) :
  HasSum (fun k => phi ((2*k+1 : Real) / 2) * (2*k+1)^(2*m)) 0

-- Exact folded DFT / alias identity.
theorem dyadicDCT_eq_aliasSum
  (n r : Nat) (hn : 0 < n) (hr : Odd r) (hrN : r < 2^n) :
  dyadicDCT n r = tsum (fun q : Int => phi (q * 2^n + r / 2))

-- Uniform certified recovery.
theorem dyadicDCT_sub_halfInteger_le
  (n r : Nat) (hn : 2 <= n) (hr : Odd r) (hrN : r < 2^n) :
  abs (dyadicDCT n r - phi (r / 2)) <= aliasError n

-- Rational Galois power trace.
theorem foldedPowerTrace_mem_rat
  (n m : Nat) : IsRat (sumOddResidues n (fun r => (dyadicDCT n r)^m))
```

# 15 Further frontier problems

## 15.1 Exact signs of the folded coefficients

[conjecture 11.4](#) is the most immediate problem. A proof would turn the finite dyadic DCT into a second exact occurrence of the Thue–Morse sign word, not merely an approximation to it. Useful routes include:

1. a quasi-periodic Green-kernel representation of  $\sum_q \Phi(q + \alpha)/(q + \alpha)^n$ ;
2. variation-diminishing or total-positivity properties of dyadic samples of up;
3. a recursion in  $n$  for the quotient  $A_{n,r}/\Gamma_{n,r}$ ;
4. an integral representation with an explicitly positive kernel.

## 15.2 Minimal polynomials and stabilizers

**Problem 15.1.** *Determine when  $\mathcal{P}_n$  is the minimal polynomial of  $A_{n,1}$  and when it is a proper power. More generally, classify all coincidences  $A_{n,r} = A_{n,s}$  and determine the discriminant and local factorization of the orbit polynomial.*

The examples  $n = 2, 3$  are irreducible over  $\mathbb{Q}$ . The exact Galois action makes this a finite arithmetic problem at each level, but no general primitivity criterion is presently known.

## 15.3 Arithmetic nature of the limiting amplitudes

**Problem 15.2.** *Prove irrationality, transcendence, or nontrivial algebraic independence results for*

$$\Phi(1/2), \Phi(3/2), \Phi(5/2), \dots$$

The cyclotomic approximants (45) come with unusually strong error bounds, but their algebraic degrees and heights also grow. Determine whether the approximation quality can beat the relevant Liouville-type height thresholds after sharp height estimates are included.

### 15.4 Spectral moments and nonlinear functionals of $\mathfrak{u}_p$

The trace limits (66) define constants  $\mathfrak{s}_m = \sum_k a_k^m$ . The cases  $m \geq 3$  do not have an immediate ordinary Parseval interpretation.

**Problem 15.3.** Find direct integral, probabilistic, or operator-theoretic representations of  $\mathfrak{s}_m$  for  $m \geq 3$ . Determine whether the rational trace sequences  $p_m(n)$  satisfy recurrences or congruences inherited from the dyadic sample array.

### 15.5 A genuine transform between the two Gaussian-binomial webs

Formula (45) composes the exact dyadic-value web with a cyclotomic DCT. The spectral determinant report constructs a different family of Gaussian-binomial coefficients from finite  $q$ -Euler determinants.

**Problem 15.4.** Construct an intrinsic matrix or incidence-algebra transform that takes the Gaussian-binomial coefficients governing exact Fabius dyadic values directly to the Gaussian-binomial coefficients governing the spectral  $q$ -Euler determinants. The DCT developed here supplies an explicit intermediate layer, but not yet a closed transform between the two coefficient arrays.

### 15.6 Lambert spectral localization

**Conjecture 15.5** (Lower-Lambert frequency localization). Let  $x \downarrow 0$  and put  $r(x) = \lambda(x)/x$  as in (9). There should exist windows  $\mathcal{W}_x$  centered on odd frequencies of order  $r(x)$ , with relative width tending to zero on the logarithmic scale, such that the exact signed sum (33) is asymptotically unchanged when restricted to  $\mathcal{W}_x$ , after an appropriate multiscale grouping that preserves Thue–Morse cancellation.

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A naive termwise truncation is unlikely to work: the all-order moment identities show that cancellations across scales are essential. A successful formulation may require dyadic blocks or a Mellin transform of the signed spectral measure (21).

### 15.7 A two-variable transform

The corrected endpoint asymptotic is controlled by poles on the vertical lattice  $2\pi i\mathbb{Z}/\log 2$ . The dyadic spectral tower has the same scale lattice, and the folded DCT coefficients are finite projections of its half-integer spectrum.

**Problem 15.6.** Construct a meromorphic two-variable transform whose residues simultaneously produce

1. the Gamma–zeta periodic endpoint function  $\Psi$ ;
2. the rational cyclotomic traces  $p_m(n)$ ;
3. the weighted Prouhet moment annihilation;
4. the log-periodic heat trace of the dyadic spectral determinant.

Such a transform would unify four presently separate manifestations of dyadic self-similarity.

## 16 Conclusion

The half-integer values of the Rvachev sinc product sit at a previously underdeveloped junction of the repository's main themes. They are at once the nonzero Fourier coefficients of the two-periodic up-function, the leading constants in every integer-zero jet, and a Thue–Morse-signed sequence. The present report shows that these roles are not merely parallel observations.

Flatness at the center forces the sequence to solve an infinite positive-weight Prouhet problem of every order. Dyadic sampling folds the sequence into exact finite discrete cosine transforms whose entries are rational Fabius values. The finite  $q = 1/2$  Gaussian-binomial formula then turns each folded coefficient into a completely explicit  $q$ -binomial–Thue–Morse–cyclotomic number. An elementary selected-factor estimate unfolds those coefficients with a uniform superexponential error. Galois conjugacy supplies rational orbit polynomials and power traces, and those traces converge to nonlinear moments of the original half-integer spectrum. Splitting the first dyadic sinc factors reveals a finite root-of-unity Thue–Morse product as the sign carrier of every alias class. Finally, the all-order Prouhet cancellation rewrites the Fabius endpoint as an exact spectral remainder beyond every algebraic order; the repository's corrected Lambert– $W$  theorem then evaluates that same signed remainder asymptotically.

The strongest immediate next theorem would be positivity of the residual shifted spectral sum (78). Beyond it lie arithmetic questions about the cyclotomic tower, transcendence of the limiting amplitudes, and a true lower-Lambert localization theorem. The finite identities and error bounds developed here are sufficiently elementary to support a staged Lean formalization before those deeper questions are settled.

## A Corpus inventory

The same-day recursive audit snapshot counted fifty-seven TeX files: twenty-five in the living or imported documentation stratum and thirty-two frozen historical versions. The current canonical frontier source is maintained by consolidation, so capitalization and the continued presence of processed draft paths can change after the snapshot. The list below records the path relative to `Analysis/FabiusFunction/docs`. Historical items are grouped by the archive directory; filenames follow the base name shown.

### A.1 Living documentation: 25 TeX files

| No. | Relative path                                                                                                                             |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | <code>FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex</code>                                                |
| 2   | <code>Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex</code>                                                            |
| 3   | <code>Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex</code>                                          |
| 4   | <code>fabius_lean_walkthrough/fabius_lean_walkthrough.tex</code>                                                                          |
| 5   | <code>non-formalized-research-frontiers/non-formalized-research-frontiers.tex</code>                                                      |
| 6   | <code>non-formalized-research-frontiers/drafts/Fabius_Thue_Morse_Formula_Atlas/Fabius_Thue_Morse_Formula_Atlas.tex</code>                 |
| 7   | <code>non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex</code>    |
| 8   | <code>non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex</code> |

| No. | Relative path                                                                                                                                                                                       |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9   | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex                                         |
| 10  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex                                                   |
| 11  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex                                         |
| 12  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex |
| 13  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex                                                                           |
| 14  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex                                                                           |
| 15  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/Rvachev_Up_Fourier_Decay_Final.tex                                                                     |
| 16  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/Rvachev_Up_Fourier_Decay_Final_Synthesis.tex                                                           |
| 17  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay_final.tex                                                                     |
| 18  | non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay_final.tex                                                                     |
| 19  | papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex                                                                                                                                                |
| 20  | papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex                                                                                                                                      |
| 21  | papers/arXiv-1502.06738v2/Lacunary_products.tex                                                                                                                                                     |
| 22  | papers/arXiv-1702.05442v1/09-Function.tex                                                                                                                                                           |
| 23  | papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex                                                                                                                                                     |
| 24  | papers/arXiv-2211.13570v1/document.tex                                                                                                                                                              |
| 25  | papers/ubiq15/ubiq15-corrected.tex                                                                                                                                                                  |

## A.2 Frozen archive: 32 TeX files

| No. | Relative path under archive/                                                                                  |
|-----|---------------------------------------------------------------------------------------------------------------|
| 1   | early-notes/FabiusAsymptotic/FabiusAsymptotic.tex                                                             |
| 2   | early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex |
| 3   | standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex        |
| 4   | standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex                                  |
| 5   | standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex                    |

| No. | Relative path under archive/                                                                                                         |
|-----|--------------------------------------------------------------------------------------------------------------------------------------|
| 6   | primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex                                                 |
| 7   | primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex                                               |
| 8   | primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex                                               |
| 9   | primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex                                                                 |
| 10  | primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex                                                                   |
| 11  | primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex                                               |
| 12  | primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex                                                  |
| 13  | primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex                                                |
| 14  | q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex                                                                 |
| 15  | q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex                                                                 |
| 16  | lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex                                               |
| 17  | lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex                                                                 |
| 18  | glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex                                               |
| 19  | glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex                                             |
| 20  | research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex                                               |
| 21  | research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex |
| 22  | research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex                                   |
| 23  | research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex                                                       |
| 24  | research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex                                         |
| 25  | research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex                                       |
| 26  | research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex                                               |
| 27  | research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex                                       |
| 28  | research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex                                     |
| 29  | research-frontiers/non-formalized-research-frontiers/non-formalized-research-frontiers.tex                                           |

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| No. | Relative path under archive/                                                                             |
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| 30  | research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex                               |
| 31  | research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex |
| 32  | research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex                               |

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## B Selected source map

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